

A FILMHEDGE + MEDIAHEDGE / RESEARCH PAPER

Divergent Correlation: Credit Structure, Lender Performance and Box Office Outcomes

A statistical and probabilistic study of debt-financed theatrical film outcomes

CONTACT

Produced by the team at FilmHedge (a Media Hedge company). Visit us on the web at <http://filmhedge.com>. If you are lender, banker, or analyst who has data you'd like to contribute to a study like this, feel free to email us at data@filmhedge.com

Abstract

This study evaluates whether senior lending credit classifications, loan structure, distributor/obligor gradings, funding channel, and gap/estimate exposure can predict market outcomes. The design uses an anonymized sample loan data set and canonical post-release performance signals and restricts each inference to projects with the most complete data required to answer the question.

The resulting evidence base contains 34 funded projects, 22 observed payoffs, 8 active loans, 4 loans with undetermined status, 19 projects with public theatrical gross, and 15 projects with both observed payoff and public gross theatrical receipts.

Rank correlations with public-gross efficiency are weak and interval estimates were broad. The unadjusted association between legacy grade and payoff hazard disappears after controlling for origination vintage: the hazard ratio changes from 1.61 (95% *credible interval* 1.03–2.37) to 1.02 (0.63–1.59). The blended-loan estimate changes direction under the same adjustment. Funding fields and the institutional indicator are structurally confounded with vintage. The current obligor methodology can be matched exactly to only 9 loans, and only 6 also have public gross, which prevents stable primary inference.

One association survives vintage adjustment: higher log LTV corresponds to faster observed payoff, hazard ratio 2.06 (1.14–3.80) per sample standard deviation. This direction runs counter to a simple leverage-risk interpretation and is best treated as a portfolio-selection or product-mix signal. It does not establish that higher leverage improves repayment but begs the question.

Public theatrical gross itself has only a weak relationship with loan payoff timing, reinforcing the separation between theatrical performance, the distributor waterfall, collateral realization, and lender repayment.

Primary Conclusion

The present data set is limited, and the sample size was relative small so the research is not adequate for guiding production underwriting coefficients at scale. The next empirical gain will come from a larger data set, cleaner longitudinal capture of realized cash flows, broader obligor entities and distributions, more variance in funding amounts, and collateral tranches.

Table of Contents

Abstract.....	2
1. Data Provenance And Cohort Construction.....	5
1.1 Source Hierarchy.....	5
1.2 Exclusions.....	6
1.3 Strong-data rule	6
1.4 Missing from the results.....	7
2. Statistical and probabilistic methods	8
2.1 Outcomes	8
2.2 Rank-based exploratory tests.....	8
2.3 Multiplicity control	8
2.4 Bayesian regression for market performance.....	8
2.5 Time-to-payoff analysis	9
2.6 Interpretation rule.....	9
3. Credit classifications and realized outcomes	10
3.1 Measurement semantics.....	10
3.2 Market performance.....	10
3.3 Payoff timing.....	10
3.4 Taxonomy alignment.....	11
4. Blended versus standalone loans	12
4.1 Nonparametric evidence	12
4.2 Model instability.....	12
5. Distributor and obligor quality	14
5.1 Empirical coverage	14
5.2 A testable future calibration target	14
6. Gap and estimate exposure.....	15
6.1 Conservative exposure definition.....	15
6.2 Time-to-payoff model	15
7. Funding architecture	16
7.1 Structural missingness.....	16
7.2 Why a funding regression would be invalid	16
8. Additional empirical questions	16
8.1 Does higher LTV predict slower payoff?	16
8.2 Does public theatrical success predict lender payoff speed?.....	17
8.3 Can the present portfolio estimate default probability?.....	17

9. External evidence and market context.....	17
10. Implications for underwriting and portfolio measurement.....	19
10.1 Underwriting implications	19
10.2 Minimum longitudinal data contract.....	19
10.3 Prospective validation design.....	19
11. Limitations	20
12. Conclusion	21
References	22
Internal Sources.....	22
External Literature and Market Research	22
Appendix A. Complete exploratory test table	23
Bayesian market-performance models	23
Bayesian payoff-hazard models	24
Vintage-adjusted payoff-hazard model	24
Appendix B. Variable definitions and reproducibility.....	25
Reproducibility record.....	25

1. Data Provenance And Cohort Construction

1.1 Source Hierarchy

We took a sample of 34 funded productions, funded by us or our partners, and then looked at their post-release performance in-market following theatrical distribution. For the **full 34-production study population**, the underwritten principle of loans was **\$5.84 million** and the mean production budgets was **\$15.04 million**. Twenty-two had observed end-to-end loan payoffs (clearing pre-distribution debt). Eight remained active, beyond the time of our study. Four had undetermined outcomes due to incomplete data. 19 had public theatrical gross and 15 had both an observed payoff and public gross. The analysis used only the records complete enough to investigate each question.

Summary of Data Population Used for this Study

	Films (n)	Total (USD)
Underwritten loan principal—all study productions	34	\$198,635,732
Underwritten loan principal—productions with reported box office	19	\$146,560,433
Reported worldwide theatrical box-office gross	19	\$144,574,865

Table 1: 34 feature films, 19 released theatrically

Study population	Measure	n	Mean (\$MM)	Median (\$MM)	SD (σ)
Full study	Loan size	34	\$5.84	\$4.81	\$5.32
Full study	Total production budget	34	\$15.04	\$14.50	\$10.77
Joint high-completeness subset ¹	Loan size	15	\$8.40	\$8.71	\$5.74
Joint high-completeness subset ¹	Total production budget	15	\$19.95	\$18.00	\$11.13

Table 2: Loan sizes and production budgets

Study population	n	Mean (\$MM)	Median (\$MM)	SD (σ)
Full study—productions with reported box office gross	19	\$7.61	\$0.75	\$15.30
Joint high-completeness subset ¹	15	\$8.93	\$0.75	\$17.03

Table 2: Productions with reported box office gross

¹ Productions with the most complete data sets across all research fields.

Only 19 of the 34 study productions reported gross theatrical income. The other 15 were excluded from these calculations, not assigned zero revenue. The distribution was strongly right skewed: three productions generated 86.3% of reported gross, raising the mean substantially above the median. In short, the box-office averages were heavily influenced by a few large results. The median describes the middle ground; while standard deviation shows how much dispersion the average alone leaves hidden. Loan amounts represent underwritten principal. Box-office gross is theatrical revenue before deductions; it does not represent lender recovery or total film revenue.

Those statistics overall describe a great variation in deal size while the box-office statistics describe variation in reported theatrical outcomes. Neither directly measures loan-default risk or any correlation between media credit underwriting and theatrical outcomes.

The portfolio contained professionally financed productions with meaningful budgets (average \$15.04 million), recognizable talent and real distribution, rather than a collection of hypothetical packages.

A normalized funded sample population of films served as the authoritative transaction-level source for principal, collateral, pricing, recorded grade and tier, status, dates, and captured funding. The canonical Post-Release Performance data (pulled from IMDB.com) supplied estimates of public worldwide theatrical gross. Internal credit grading mapped to their normal rate-implied pricing bands. Our distributor database analyzed 153 film distributors, domestic and global. Source lineage is recorded in an internal private manifest to preserve the identity of our partners, producers and customers.

1.2 Exclusions

No title, alias, distributor-linked project name, or plot-level identifier appears in the analytic dataset, exhibits, result files, or paper. A separate private ledger preserves the one-to-one crosswalk for authorized use.

1.3 Strong-data rule

A universal complete-case filter would discard useful observations and would still fail to solve outcome-specific missingness. The study therefore applies a question-specific strong-data rule: an observation enters a model only when the exposure, the relevant outcome, the timing or censoring indicator, and the essential scale denominator are observed and internally valid. Outcome fields are never imputed.

$C_i = K^{-1} \sum_k I(x_{ik} \text{ is observed and valid})$ C_i is the descriptive completeness score for project i across K core fields; $I(\cdot)$ is an indicator. Cohort membership still follows explicit field requirements rather than a single arbitrary threshold.

Cohort	Inclusion rule	<i>n</i>	Analysis
Study base	Normalized funded loan; requested exclusions removed	34	Portfolio description and exposure prevalence
Resolved payoff	Status paid off; actual payoff date observed	22	Payoff-day rank and group comparisons
Primary survival	Paid off or active; valid follow-up; undetermined excluded	30 (22 events)	Kaplan–Meier and Weibull proportional hazards
Public gross	Worldwide public gross and positive budget observed	19	Gross-to-budget models
Joint high-completeness	Paid off plus public gross, valid budget and principal	15	Cross-domain lender/market tests
Current obligor matched	Exact approved entity match	9; 6 with gross	Descriptive only; primary model not estimable

Source: anonymized study dataset

1.4 Missing from the results

Availability is patterned by loan status and vintage. Public gross is concentrated among historical paid-off projects. Funding detail is concentrated among current active loans. Gap/estimate labels are concentrated among unresolved observations. These patterns violate the assumptions required for naïve complete-case regression to represent the full portfolio. The analysis treats missingness as a measurement mechanism and limits the scope of each claim accordingly.



Figure 2. Data availability is structured by funded status.

2. Statistical and probabilistic methods

2.1 Outcomes

Lender performance is represented primarily by time from disbursement to observed payoff. Active loans contribute follow-up time and are skewed at the analysis date. The normalized resolved cohort contains zero borrower defaults, so a default-probability regression is unidentified: an all-zero response contains no information about covariate separation. Public worldwide theatrical gross is used only as a market-performance proxy and is scaled by documented budget (pulled from underwriting data, not IMDB.com which can over or underestimate cost). The data does not measure net film revenue, distributor receipts, collateral collections, or lender cash recovery.

$$G_i = \text{Gross}_i / \text{Budget}_i; \quad Y_i = \ln(G_i)$$

G_i is public-gross efficiency. The natural logarithm Y_i reduces the leverage of the film market's heavy right tail and converts multiplicative effects into additive coefficients.

2.2 Rank-based exploratory tests

Monotone associations are summarized with Spearman rank correlation because sample sizes are small, rating scales are ordinal, and gross efficiency is strongly skewed. Group comparisons use median differences and Cliff's delta. Confidence intervals use 20,000 nonparametric bootstrap resamples. Two-sided permutation p-values use 20,000 label or rank permutations under the null.

$$\rho_s = \text{corr}(\text{rank}(X), \text{rank}(Y))$$

Spearman's ρ_s measures monotone association without assuming linearity or Gaussian errors. A value near zero indicates weak rank ordering, not proof of independence.

$$\hat{p}^* = [1 + \sum_{\beta} I(|T_{\beta}| \geq |T_{\text{obs}}|)] / (B + 1)$$

The finite-simulation correction yields a valid permutation *p-value* when $B = 20,000$ random null reallocations are used.

2.3 Multiplicity control

Ten pre-specified exploratory tests are evaluated together with the Benjamini-Hochberg procedure. This controls the expected false discovery rate among rejected hypotheses under the standard dependence conditions. The smallest adjusted *q-value* is approximately 0.854; no exploratory association survives multiplicity correction.

$$q(i) = \min_{j \geq i} \{ m p(j) / j \}, \text{ with monotone truncation at } 1$$

The ordered p-values $p(j)$ are multiplied by the number m of tested hypotheses and divided by rank j ; reverse cumulative minima enforce monotonicity.

2.4 Bayesian regression for market performance

Separate small-sample Bayesian linear models estimate the relationship between each exposure and log gross-to-budget. Continuous predictors are centered and scaled by their sample standard deviation; binary predictors are coded 0/1. Weakly informative Gaussian priors regularize coefficients while leaving the observed likelihood dominant at plausible effect sizes. Posterior effects are exponentiated to report multiplicative ratios in gross efficiency.

$$Y_i = \alpha + \beta X_i + \varepsilon_i, \quad \varepsilon_i \sim \text{Normal}(0, \sigma^2); \quad \exp(\beta) = \text{multiplicative outcome ratio}$$

For a standardized continuous X , $\exp(\beta)$ is the expected gross-efficiency ratio per one sample standard deviation. For a binary X , it is the expected ratio for group 1 relative to group 0.

2.5 Time-to-payoff analysis

Kaplan-Meier curves estimate the probability that a loan remains unpaid after disbursement. The 730-day restricted mean time unpaid (RMST) integrates each survival curve through two years and provides an interpretable time-scale contrast even when median payoff is unstable. Bayesian Weibull proportional-hazards models estimate covariate effects with right censoring. Hazard ratios above one mean faster payoff, not greater default risk.

$$\hat{S}(t) = \prod_{t_j \leq t} (1 - d_j/n_j)$$

The Kaplan-Meier estimator multiplies conditional survival factors at each payoff time t_j , with d_j payoffs among n_j loans at risk.

$$\text{RMST}(\tau) = \int_0^\tau \hat{S}(t) dt$$

At $\tau = 730$ days, RMST is the expected time a loan remains unpaid during the first two years after disbursement.

$$h_i(t) = k t^{k-1} \exp(\alpha + x_i' \beta); \quad S_i(t) = \exp[-t^k \exp(\alpha + x_i' \beta)]$$

The Weibull model supplies a parametric baseline hazard and multiplicative covariate effects $\exp(\beta)$. The multivariable specification includes legacy grade, blended structure, log LTV, and origination vintage.

Posterior simulation used four Markov chains. The largest split-chain potential scale reduction statistic was $\hat{R} = 1.008$, below the conventional 1.01 diagnostic threshold. This supports convergence of the implemented sampler; it does not cure limited information in the data.

2.6 Interpretation rule

An association is treated as decision-relevant only when its interval excludes a practically neutral effect, its direction persists across reasonable specifications, its exposure has adequate within-cohort variation, and the measurement process does not mechanically confound it with vintage or status. Posterior probabilities are reported as graded evidence. They are not converted into binary declarations when credible intervals remain wide.

3. Credit classifications and realized outcomes

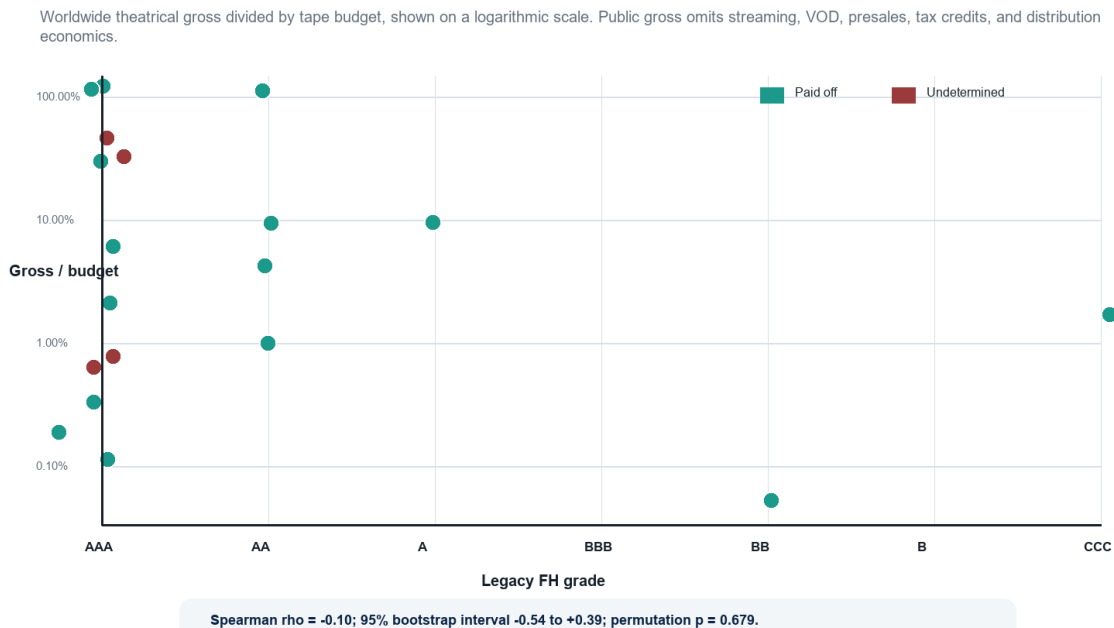
3.1 Measurement semantics

Three concepts must remain separate. Legacy FH Grade is an inherited ordinal credit field. Recorded portfolio tier is another legacy tape field. Current MH Score and Tier are pricing classifications derived from approved blended-rate bands. Because the current score is mechanically implied by price, it is not an independent ex ante covariate and cannot validate itself through a price-linked regression. Risk Score and Obligor Score are distinct underwriting concepts. Treating these fields as interchangeable would create a category error.

For descriptive comparison, lower legacy grade-risk codes represent stronger grades and lower tier numbers represent stronger tiers. The direction of any coefficient must therefore be interpreted with care. A positive payoff hazard ratio for grade risk would mean weaker recorded grades paid faster, not that risk was lower.

3.2 Market performance

Among the 19 public-gross observations, legacy grade has *Spearman* $\rho = -0.10$ (95% bootstrap interval -0.54 to 0.39; permutation $p = 0.679$). Recorded tier has $\rho = -0.21$ (-0.62 to 0.27; $p = 0.391$). The corresponding Bayesian models estimate gross-efficiency ratios of 0.57 (0.20–1.65) per one-standard-deviation worsening in legacy grade and 0.48 (0.17–1.33) for recorded tier. Both intervals include large benefit and harm.



Anonymized MediaHedge portfolio sample | Analysis date: 1 September 2026

3.3 Payoff timing

The resolved-only rank test yields $\rho = -0.16$ for legacy grade versus payoff days and $\rho = -0.28$ for recorded tier; both intervals span zero. A univariable survival model produces an apparently faster payoff hazard for weaker legacy grade, *HR* 1.61 (1.03–2.37). Once origination vintage, blended

structure, and log LTV are included, the grade estimate collapses to *HR 1.02 (0.63–1.59)*. The posterior probability that the adjusted HR exceeds one is 0.538, effectively equivocal.

3.4 Taxonomy alignment

Among 29 loans whose rate falls inside a current published band, recorded portfolio tier and current price-implied credit grading agree exactly for 4 observations (13.8%). Quadratic weighted Cohen's κ is -0.35. Five additional rates occupy uncovered intervals. This is evidence that the fields encode different taxonomies, not evidence that either taxonomy is intrinsically wrong.

	T1	T2	T3	T4	T5
Tier 1	1	2	6	0	0
Tier 2	6	2	5	0	1
Tier 3 Recorded portfolio tier	0	1	1	0	0
Tier 4	3	1	0	0	0
Tier 5	0	0	0	0	0

Exact agreement: 4 of 29 (13.8%). Quadratic weighted kappa = -0.35. This is evidence against treating the fields as interchangeable, not evidence that either taxonomy is intrinsically wrong.

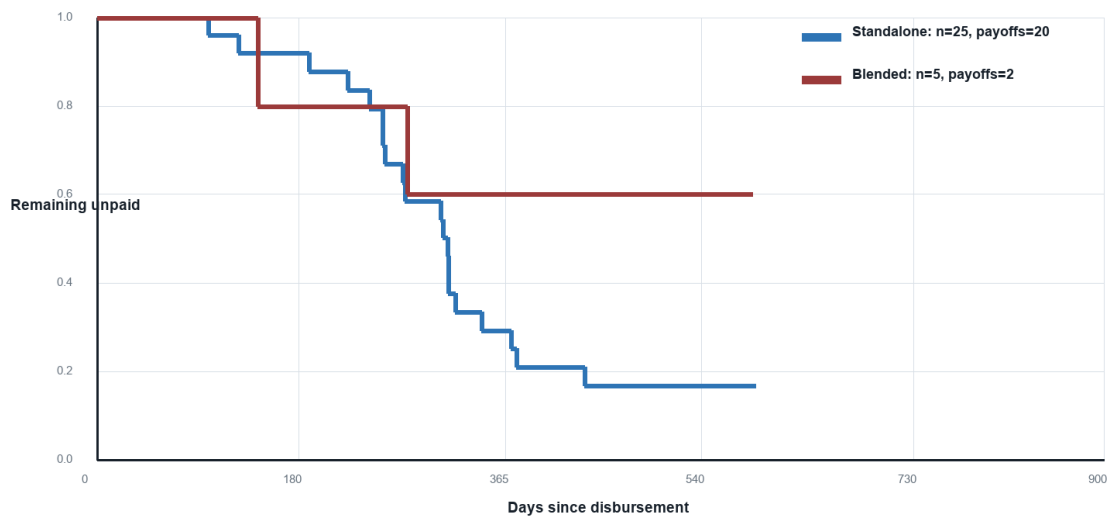
Anonymized MediaHedge portfolio sample | Analysis date: 1 September 2026

4. Blended versus standalone loans

4.1 Nonparametric evidence

Five of the 30 loans in the primary survival cohort are blended, but only two have observed payoff events. At 730 days, the Kaplan-Meier restricted mean time unpaid is 166 days longer for blended loans, with a 95% bootstrap interval from -82 to 393 days. Among the 22 paid-off loans, the median payoff-day contrast points in the opposite direction: blended minus standalone = -81 days (-169 to 20; $p = 0.193$). The discrepancy is expected because the resolved-only comparison conditions on eventual payoff and discards active censoring. Different collateral types pay out at different times so this is consistent with blended loans where, say the tax credit pays early, but pre-sales are delayed, or vice versa.

Blended loans show slower observed payoff, with wide uncertainty.



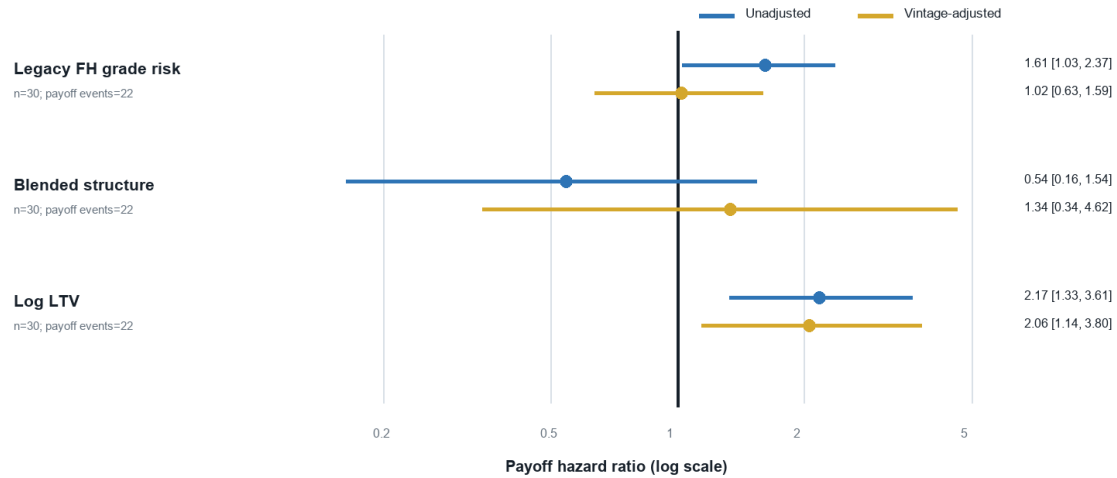
Anonymized MediaHedge portfolio sample | Analysis date: 1 September 2026

4.2 Model instability

The univariable Bayesian Weibull model estimates $HR\ 0.54\ (0.16-1.54)$, consistent with slower payoff. The vintage-adjusted model estimates $HR\ 1.34\ (0.34-4.62)$, changing direction. In the public-gross cohort, the Bayesian gross-efficiency ratio is 11.4, but its 95% credible interval spans 0.45 to 297.1; only two blended observations supply that contrast. The high posterior probability of a positive gross coefficient is therefore driven by sparse, extreme observations and cannot support a general claim. Our access to data is too few as a young company, but over time we expect the sample integrity to improve.

Rating and structure effects are unstable after vintage adjustment.

Bayesian Weibull proportional-hazards models. HR > 1 indicates faster payoff; grade and LTV are scaled per one sample standard deviation.



The apparent unadjusted grade signal disappears after adding origination vintage. The blended estimate changes direction. LTV retains an inverse-to-risk association, consistent with product-mix or selection effects rather than a causal benefit of higher leverage.

Anonymized MediaHedge portfolio sample | Analysis date: 1 September 2026

Conclusion on blended loans

Evidence is inconclusive. Blended loans do not show a stable advantage or disadvantage in payoff timing or public-gross efficiency. The current sample contains too few blended payoff events to separate structure from origination vintage, collateral mix, and selection into blended products. From a controls and leverage perspective, they may offer strategic advantages that cannot be proofed by this study.

5. Distributor and obligor quality

5.1 Empirical coverage

Nine loans appeared in the 34-project study base. All nine are recorded as paid off. Six also have public gross: five are rated AA and one is BB under the current obligor scale. This creates three barriers to regression: a tiny sample, almost no within-score replication, and no matched loan with an adverse resolved lender outcome. A fitted coefficient would be determined by individual records rather than a reproducible portfolio relationship. The primary obligor model is therefore declared non-estimable.

Conclusion on obligor/distributor quality

The current matched sample is directionally consistent with strong obligors appearing in successful payoffs, but it contains no valid counterfactual. The analysis cannot establish calibration, rank ordering, or predictive lift.

5.2 A testable future calibration target

Once sufficient outcomes accumulate, the preferred target is not theatrical gross. It is realized collateral collection or time to lender payoff, measured by obligor and instrument. A hierarchical model can then partially pool sparse obligors within rating bands while retaining entity-level variation:

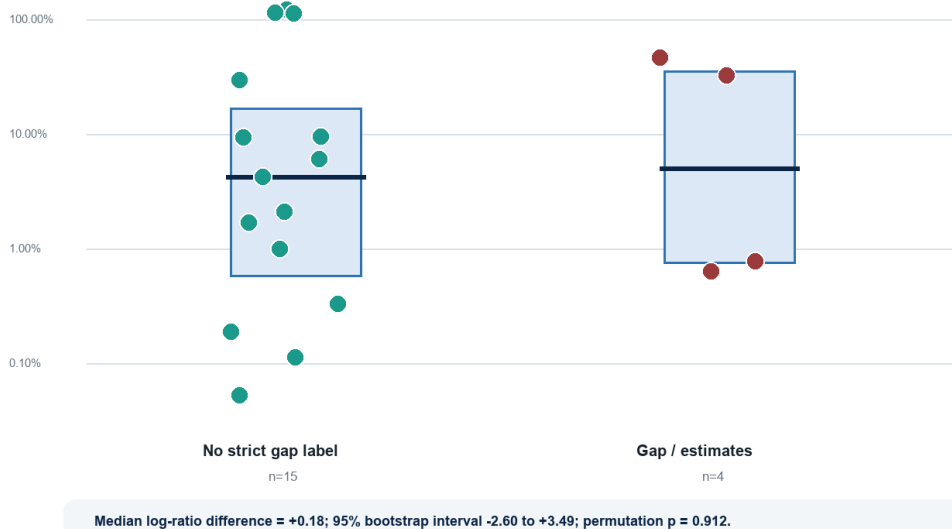
$\log h_i(t) = \log h_d(t) + \beta \cdot \text{Score}_i + u_j(\text{obligor}_i)$, $u_j \sim \text{Normal}(0, \tau^2)$ β estimates the rating-scale gradient and u_j captures repeated-obligor deviation. This model becomes identifiable only after multiple obligors, multiple ratings, and both favorable and adverse outcomes are observed.

6. Gap and estimate exposure

6.1 Conservative exposure definition

Gap/estimate exposure is defined strictly from the current normalized collateral label. Historical notes and ambiguous collateral descriptions are not re-coded into the primary variable. Eight of 34 loans carry the gap label. Only one is in the paid-off resolved cohort, four have public gross, and none belongs to the 15-project joint cohort. All four gross-observed gap/estimate loans have undetermined tape status.

Within the 19 public-gross observations, the median log gross-efficiency difference is 0.18 (-2.60 to 3.49; *permutation p* = 0.912; *Cliff's delta* = 0.07). The Bayesian multiplicative ratio is 1.47, with a 95% credible interval from 0.11 to 18.92. The interval spans roughly a 90% reduction to a nineteen-fold increase.



Anonymized MediaHedge portfolio sample | Analysis date: 1 September 2026

Gap/estimate exposure is currently inseparable from unresolved status.

6.2 Time-to-payoff model

The univariable survival model estimates a payoff *HR* of 0.41 (0.11–1.20), with posterior probability 0.948 that the *HR* is below one. The interval includes the neutral value and the exposure is correlated with unresolved status and recent vintage. Adding it to a saturated small-sample multivariable model would trade visible confounding for unstable parameter estimation.

Conclusion on gap/estimates

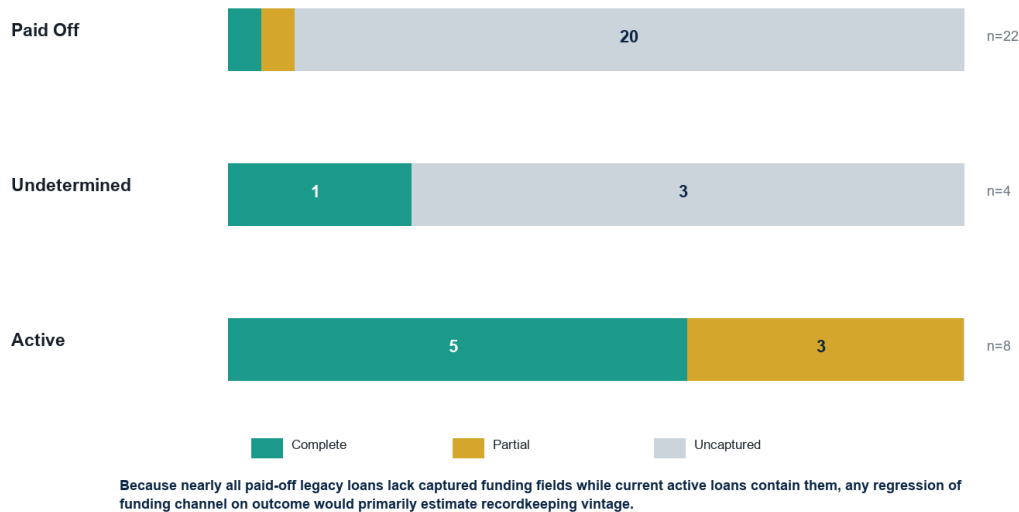
The present data cannot determine whether gap/estimate lending improves or impairs lender performance. Public theatrical gross does not answer the lender question, and the strict exposure is almost absent from resolved cases.

7. Funding architecture

7.1 Structural missingness

Only 7 of 34 loans have captured cash plus forward-flow amounts that reconcile to underwritten principal. Eleven have any positive captured funding amount. Among 22 paid-off loans, only 1 is had complete data; among 8 active loans, 5 had complete data sets. Missing or zero capture should not be interpreted as a zero economic contribution, here it meant market reporting lagged behind the timing required to be included in this study.

Funding-source completeness is confounded with vintage and outcome.



Anonymized MediaHedge portfolio sample | Analysis date: 1 September 2026

7.2 Why a funding regression would be invalid

The institutional indicator yields an unadjusted payoff *HR* of 7.07 (2.25–24.82). Every observed payoff in the primary survival cohort is marked institutional, whereas several active loans are not. This apparent effect is a historical classification boundary: the model predicts cohort membership and data-vintage practice. It does not identify a causal funding-channel benefit.

8. Additional empirical questions

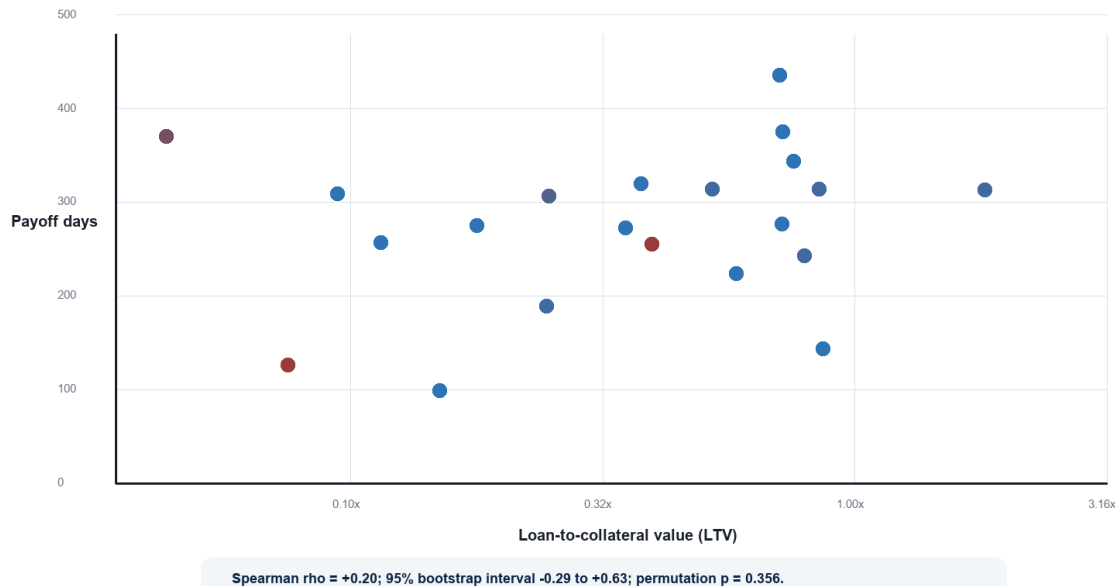
8.1 Does higher LTV predict slower payoff?

Among resolved loans, LTV has a weak positive rank correlation with payoff days, $p = 0.20$ (–0.29 to 0.63; $p = 0.356$). In the censored survival model, however, log LTV produces *HR* 2.17 (1.33–3.61) unadjusted and *HR* 2.06 (1.14–3.80) after vintage adjustment. The posterior probability that the adjusted *HR* exceeds one is 0.991.

This inverse-to-risk association is statistically persistent in the fitted model but causally ambiguous. Likely mechanisms include selective approval of high-LTV loans with stronger collateral certainty, different collateral composition, shorter effective collection cycles, or underwriting that allows higher

leverage only when hidden risk controls are favorable. The result should trigger a product-mix audit, not a higher-LTV policy.

LTV has no stable monotonic relationship with payoff time among resolved loans.



8.2 Does public theatrical success predict lender payoff speed?

In the 15-project joint cohort, gross efficiency and payoff days have Spearman $\rho = -0.16$ (-0.62 to 0.47; $p = 0.556$). The weak association is substantively important. Media financiers lend against defined collateral and cash-flow rights; theatrical gross is upstream of distributor fees, exhibitor splits, territory accounting, presales, tax credits, streaming and VOD revenue, and collection timing. A film can be theatrically weak yet repay secured collateral, or theatrically strong while the lender waits on documentation and waterfall realization.

8.3 Can the present portfolio estimate default probability?

No. The resolved cohort contains zero coded default events. Logistic regression, penalized logistic regression, and a score-calibration curve cannot recover an empirical covariate gradient from an outcome with no variation. A Bayesian intercept-only model could produce a prior-sensitive upper bound, but it would answer a different question and could be misread as portfolio calibration. The responsible result is that default-risk modeling is unidentified until default, restructuring, impairment, or realized-loss states are recorded with precise definitions.

9. External evidence and market context

The portfolio findings are consistent with the broader filmed-entertainment evidence base. De Vany and Walls characterize motion-picture outcomes as highly uncertain and heavy-tailed, with a small number of extreme successes dominating aggregate results. This supports log scaling, rank statistics, and resistance to extrapolating from a few high-gross observations. Hennig-Thurau and coauthors document how the digital era makes success measurement more complex while detailed performance

data remain inaccessible, particularly across streaming and ancillary channels. Public box office offers an incomplete proxy for ultimate film economics ('ultimates').

European Audiovisual Observatory studies also show that film-financing structures vary materially across markets and combine direct public funding, incentives, presales, broadcaster investment, and other sources. Its recent work on incentives emphasizes timing, eligibility, administrative complexity, and predictability—dimensions that can affect collection duration independently of headline collateral amount.

The methodological choices also follow established statistical practice. Riley and colleagues show that prediction-model sample size must be driven by overfitting control and precision rather than a simplistic events-per-variable rule. Weakly informative priors, as developed by Gelman and colleagues for regression, stabilize sparse estimates without manufacturing information. Benjamini and Hochberg supply the false-discovery-rate correction used for exploratory tests, and Kaplan and Meier supply the product-limit estimator used for censored payoff times.

10. Implications for underwriting and portfolio measurement

10.1 Underwriting implications

1. Keep pricing classifications, risk scores, obligor scores, and legacy tape labels as separate versioned variables. Do not backfill one taxonomy from another.
2. Do not adjust approval thresholds based on the observed inverse LTV association. Audit the selection mechanism and collateral mix first.
3. Do not infer a blended-structure premium or penalty. Require additional resolved blended events and tranche-level collateral data.
4. Treat gap/estimate exposure as a structured collateral position with amount, priority, estimate source, haircut, and collection path—not as a binary project label.
5. Evaluate obligor quality against realized collections and payoff timing, not public gross alone.

10.2 Minimum longitudinal data contract

Domain	Required fields	Statistical purpose
Identity	Immutable project_id, loan_id, tranche_id	Prevents title-based joins and supports repeat facilities
Score versioning	score_name, score_value, scale, version, as_of_date, analyst	Separates legacy, price-implied, risk, and obligor constructs
Obligor	legal entity, role, territory, guide record ID, rating as of origination	Enables exact joins and prospective calibration
Collateral	type, face amount, haircut, priority, contract, estimate source, expected date	Separates gap, estimates, presales, tax credits, MGs, and other rights
Funding	source, committed amount, funded amount, funding date, tranche allocation	Distinguishes proposed, committed, and funded capital
Cash realization	receipt date, source, gross receipt, fees, net lender cash, currency	Creates the correct lender-performance outcome
Status	active, paid off, default, restructured, impaired, undetermined; effective date	Provides valid censoring and adverse-event definitions
Market performance	theatrical, streaming license, VOD, presales, ancillary, territory	Reduces dependence on public box office
Audit fields	source record, ingestion date, data owner, correction history	Makes missingness and revisions observable

Recommended prospective data contract derived from the identification failures observed in this study.

10.3 Prospective validation design

1. Freeze every origination-time exposure before outcome observation, including score version and obligor rating.
2. Record monthly cash-flow and status snapshots so that active follow-up, restructuring, and partial recovery are explicit.

3. Pre-register the primary endpoint as time to final payoff or discounted recovery ratio; keep public gross secondary.
4. Use temporal validation: train on earlier vintages and evaluate calibration and discrimination on later vintages.
5. Bootstrap the entire modeling pipeline, including feature selection and calibration, until a truly external validation cohort exists.
6. Report calibration-in-the-large, calibration slope, time-dependent Brier score, concordance, and decision-curve net benefit. A rank metric alone cannot validate underwriting economics.

11. Limitations

Limitation	Observed condition	Consequence
Sample size	34 total; 22 payoff events; 19 gross observations	Large effects can remain compatible with the data; multivariable flexibility is sharply limited
No resolved defaults	Zero coded defaults	Default probability and score calibration cannot be estimated
Outcome proxy	Public theatrical gross only	Omits streaming, VOD, presales, incentives, fees, waterfalls, and lender cash
Right censoring	Eight active loans	Recent vintages have less time to pay off; survival analysis is required
Undetermined status	Four loans	Excluded from primary survival analysis; cannot be presumed active or paid off
Funding capture	Complete for seven loans	Channel comparisons are confounded by documentation vintage
Obligor linkage	Nine exact matches	No stable score gradient or adverse counterfactual
Taxonomy drift	Legacy and current tier systems disagree	Historical labels cannot be pooled without versioned semantics
Observational design	Underwriting selects exposures	Coefficients are associations, not causal treatment effects

12. Conclusion

The analyzed portfolio does not currently demonstrate that recorded credit ratings, blended structure, obligor/distributor ratings, funding channel, or gap/estimate exposure independently determine ultimate market performance or payoff timing.

The apparent rating effect on payoff is eliminated by origination-vintage adjustment. The blended effect changes direction. Obligor and funding fields lack the coverage and counterfactual variation required for regression. Gap exposure is nearly absent among resolved loans and is fully entangled with undetermined status in the gross-observed subset.

The only adjusted association with a concentrated posterior is faster payoff at higher log LTV. Its direction contradicts a simple leverage-risk mechanism and most plausibly reflects selection, collateral composition, or product design. It is a hypothesis for targeted audit, not an underwriting rule. Public theatrical performance has weak correspondence with payoff speed, confirming that lender outcomes require direct cash-flow measurement.

MediaHedge can convert this portfolio from a descriptive record into a validated underwriting dataset by implementing versioned score semantics, exact obligor identities, tranche-level collateral and funding, dated status transitions, and realized cash receipts. With that structure in place, the next research cycle can test calibration, compare product structures prospectively, and quantify decision value. Until then, the scientifically defensible conclusion is that the requested relationships remain unproven.

References

Internal Sources

1. FilmHedge proprietary credit analytics platform, credit-rating risk, and distributor/obligor and data sets. Canonical knowledgebase, accessed 1 September 2026.
2. MediaHedge. Partner Loan Tape Normalized, 27 July 2026. Internal source workbook.

External Literature and Market Research

1. De Vany, A. S., & Walls, W. D. (1999). Uncertainty in the Movie Industry: Does Star Power Reduce the Terror of the Box Office? *Journal of Cultural Economics*, 23, 285–318. [doi:10.1023/A:1007608125988](https://doi.org/10.1023/A:1007608125988)
2. Hennig-Thurau, T., Ravid, S. A., & Sorenson, O. (2021). The Economics of Filmed Entertainment in the Digital Era. *Journal of Cultural Economics*, 45, 157–170. [doi:10.1007/s10824-021-09407-6](https://doi.org/10.1007/s10824-021-09407-6)
3. European Audiovisual Observatory. (2025). Financing fiction film production in Europe: 713-film sample and financing-source analysis. [Market research release](#)
4. European Audiovisual Observatory. (2026). Fiscal incentives and cash rebates in the audiovisual sector. [Market research report](#)
5. Riley, R. D., Ensor, J., Snell, K. I. E., et al. (2020). Calculating the sample size required for developing a clinical prediction model. *BMJ*, 368, m441. [doi:10.1136/bmj.m441](https://doi.org/10.1136/bmj.m441)
6. Gelman, A., Jakulin, A., Pittau, M. G., & Su, Y.-S. (2008). A weakly informative default prior distribution for logistic and other regression models. *Annals of Applied Statistics*, 2(4), 1360–1383. [doi:10.1214/08-AOAS191](https://doi.org/10.1214/08-AOAS191)
7. Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society, Series B*, 57(1), 289–300. [doi:10.1111/j.2517-6161.1995.tb02031.x](https://doi.org/10.1111/j.2517-6161.1995.tb02031.x)
8. Kaplan, E. L., & Meier, P. (1958). Nonparametric estimation from incomplete observations. *Journal of the American Statistical Association*, 53(282), 457–481. [doi:10.1080/01621459.1958.10501452](https://doi.org/10.1080/01621459.1958.10501452)

Appendix A. Complete exploratory test table

Question	Cohort	n	Estimate	95% interval	p	BH q
Legacy FH grade risk vs payoff days	Resolved paid-off cohort	22	-0.157	[-0.556, 0.303]	0.485	0.854
Recorded tier vs payoff days	Resolved paid-off cohort	22	-0.278	[-0.655, 0.260]	0.213	0.854
LTV vs payoff days	Resolved paid-off cohort	22	0.204	[-0.290, 0.625]	0.356	0.854
Blended vs standalone payoff days	Resolved paid-off cohort	22	-80.682	[-169.365, 20.000]	0.193	0.854
Legacy FH grade risk vs log gross-to-budget	Public-gross cohort	19	-0.100	[-0.537, 0.385]	0.679	0.854
Recorded tier vs log gross-to-budget	Public-gross cohort	19	-0.210	[-0.617, 0.273]	0.391	0.854
LTV vs log gross-to-budget	Joint high-completeness cohort	15	0.114	[-0.498, 0.649]	0.683	0.854
Loan-to-budget vs log gross-to-budget	Joint high-completeness cohort	15	0.050	[-0.564, 0.625]	0.864	0.912
Public gross efficiency vs payoff days	Joint high-completeness cohort	15	-0.163	[-0.625, 0.470]	0.556	0.854
Gap / estimates vs other collateral: log gross-to-budget	Public-gross cohort	19	0.176	[-2.604, 3.493]	0.912	0.912

Intervals are percentile bootstrap intervals. P-values are two-sided permutation values. Estimate is Spearman ρ for correlation tests and group-1-minus-group-0 median difference for group tests.

Bayesian market-performance models

Predictor	n	Gross-efficiency ratio	95% credible interval	P($\beta > 0$)
Legacy grade risk	19	0.57	[0.20, 1.65]	0.141
Recorded tier	19	0.48	[0.17, 1.33]	0.076
Gap/estimates	19	1.47	[0.11, 18.92]	0.618
Blended	19	11.40	[0.45, 297.08]	0.931
LTV	15	1.29	[0.38, 4.45]	0.667
Loan/budget	15	1.30	[0.37, 4.54]	0.670

Continuous effects are per one sample standard deviation; binary effects compare group 1 with group 0.

Bayesian payoff-hazard models

Predictor	n	Events	HR	95% credible interval	P(HR > 1)
Legacy FH grade risk	30	22	1.61	[1.03, 2.37]	0.981
Recorded portfolio tier	30	22	1.51	[1.03, 2.15]	0.983
Blended structure	30	22	0.54	[0.16, 1.54]	0.130
Gap / estimates label	30	22	0.41	[0.11, 1.20]	0.052
Log LTV	30	22	2.17	[1.33, 3.61]	1.000
Institutional indicator	30	22	7.07	[2.25, 24.82]	1.000

Univariable Weibull proportional-hazards models. HR > 1 means faster payoff.

Vintage-adjusted payoff-hazard model

Predictor	HR	95% credible interval	P(HR > 1)
Legacy FH grade risk	1.02	[0.63, 1.59]	0.538
Blended structure	1.34	[0.34, 4.62]	0.676
Log LTV	2.06	[1.14, 3.80]	0.991
Origination vintage	0.29	[0.15, 0.56]	0.000

$n = 30$; payoff events = 22; maximum R^2 = 1.008.

Appendix B. Variable definitions and reproducibility

Variable	Definition	Interpretation boundary
event_paid_off	1 for observed payoff; 0 for active right-censoring	Undetermined observations excluded from primary survival
followup_days	Disbursement to payoff or censoring date	Time scale for survival analysis
legacy_grade_risk	Ordinal encoding of legacy FH Grade	Higher means weaker recorded grade
recorded_tier	Legacy portfolio tier from normalized tape	Not current price-implied MH Tier
current_price_implied_tier	Current canonical tier band implied by approved blended rate	Descriptive taxonomy comparison only
blended	Multiple loan/collateral components under normalized structure label	Binary exposure
gap_or_estimates_strict	Current collateral label explicitly contains gap or estimate	No ambiguous historical recoding
ltv	Underwritten principal / collateral coverage	Log transformed for hazard models
gross_to_budget	Public worldwide theatrical gross / tape budget	Secondary market-performance proxy
funding_complete	Captured funding amounts reconcile to underwritten principal	Missing capture is not zero funding
obligor_primary_risk	Worst risk score among exact approved entity matches	Primary model requires exact match

Definitions reflect the analytic transformation layer; source fields remain unchanged.

Reproducibility record

Our research archive contains the anonymized datasets, data-construction scripts, analysis scripts, machine-readable results, and figures. Random seeds are fixed for bootstrap, permutation, posterior simulation, and MCMC. The public source archive excludes the private title ledger and internal source manifest. The report was generated with python-docx, rendered to page images, and visually inspected after metadata scrubbing.

Statistical estimates in this paper are generated from the machine-readable analysis summary rather than manually re-typed. This reduces transcription drift between the code, figures, tables, and narrative.

If you'd like to request the anonymized data for education or commercial purposes, please specify which and email data@filmhedge.com

Contact

If you'd like to learn more about media credit, private credit in media, media production lending (MPL), or other opportunities in the film finance space, contact our team at FilmHedge:

- Lenders and investors interested in learning more about opportunities in the space can contact: lender@filmhedge.com
- TV/Film producers can email: producer@filmhedge.com
- If you have a question about the data in this report or want to consult our data team, contact: data@filmhedge.com

For more reports and research like this, please visit: <http://filmhedge.com/reports>

Written by:

Jon Gosier, CEO, FilmHedge + MediaHedge

Phone: (770) 268-3006

Email: jon@filmhedge.com

September 3rd, 2026